

SUPERSTORE EXECUTIVE SALES ANALYTICS

End-to-End Data Pipeline & Executive Intelligence Report (2019 – 2020)

TOTAL REVENUE	TOTAL PROFIT	CLEANED ROWS	TIME HORIZON	ETL TOOL
\$1.57M	\$175.26K	~5,900 Records	2019 – 2020	Alteryx

1. PROJECT ARCHITECTURE & END-TO-END WORKFLOW

This analytics project demonstrates an end-to-end data pipeline. Raw transaction data (SuperStore_Sales_Dataset) was extracted from GitHub, cleansed and structured using Alteryx Designer processing approximately 5,900 valid data records across 2019 and 2020, and ingested into Microsoft Power BI for dynamic reporting.

2. DATA ENGINEERING & ETL PIPELINE (ALTERYX)

Alteryx Designer was utilized to execute Data Engineering and Quality Transformation workflows on the raw CSV dataset:

- **Data Extraction:** Ingested raw Superstore Sales Dataset from GitHub repository.
- **Data Cleansing & Transformation:** Filtered and cleaned 5,900 high-quality records, removed null entries, cleansed data formats, and handled duplicates.
- **Timeline Scoping:** Filtered dataset to focus on the 2-year analysis window (2019 – 2020).
- **Data Export:** Exported output table (**Cleaned_Sales_Data**) for Power BI modeling.

3. DASHBOARD ARCHITECTURE & VISUAL COMPONENTS

Visual Element	Visual Type	Key Metric / Field	Business Purpose
KPI Scorecards	Multi-row Cards	Sales (\$1.57M), Profit (\$175.26K), Quantity (22K)	High-level executive metrics at a glance
Sales by Category	Horizontal Bar Chart	Sum of Sales by Category	Compare performance across Office Supplies, Tech & Furniture
Sales by Segment	Donut Chart	Sum of Sales by Segment	Analyze market share (Consumer: 48.09%, Corporate, Home Office)
Sales Trend	Line / Area Chart	Sum of Sales by Order Date	Track historical sales trends across the daily order timeline (2019–2020)
Product Distribution	Bar / Column Chart	Count of Product Name by Sales	Examine product density across sales brackets
Interactive Controls	Slicer Tiles & Buttons	Category Slicer & Order Date Slicer	Dynamic cross-filtering across visual elements

4. STRATEGIC BUSINESS INSIGHTS (2019 – 2020 PERFORMANCE)

Key Takeaway: Consumer Segment Dominance

Out of 5,900 sales records across 2019–2020, the Consumer segment generated nearly half of overall sales volume at 48.09% (\$753K), serving as the primary revenue driver.

- **Category Leadership:** Technology and Office Supplies dominated total sales, while Furniture sub-categories showed narrower profit margins requiring pricing optimization.
- **Seasonal Dynamic:** Significant quarterly sales spikes observed during Q3 and Q4 across both 2019 and 2020 timelines.

5. TECHNICAL STACK

- **ETL / Data Engineering:** Alteryx Designer (Cleansed ~5,900 Records)
- **Data Source:** GitHub (Raw Superstore CSV Dataset)
- **Business Intelligence:** Microsoft Power BI Desktop
- **Calculations & Interactivity:** DAX Expressions, Category & Order Date Slicers, Dynamic Bookmark Reset Action